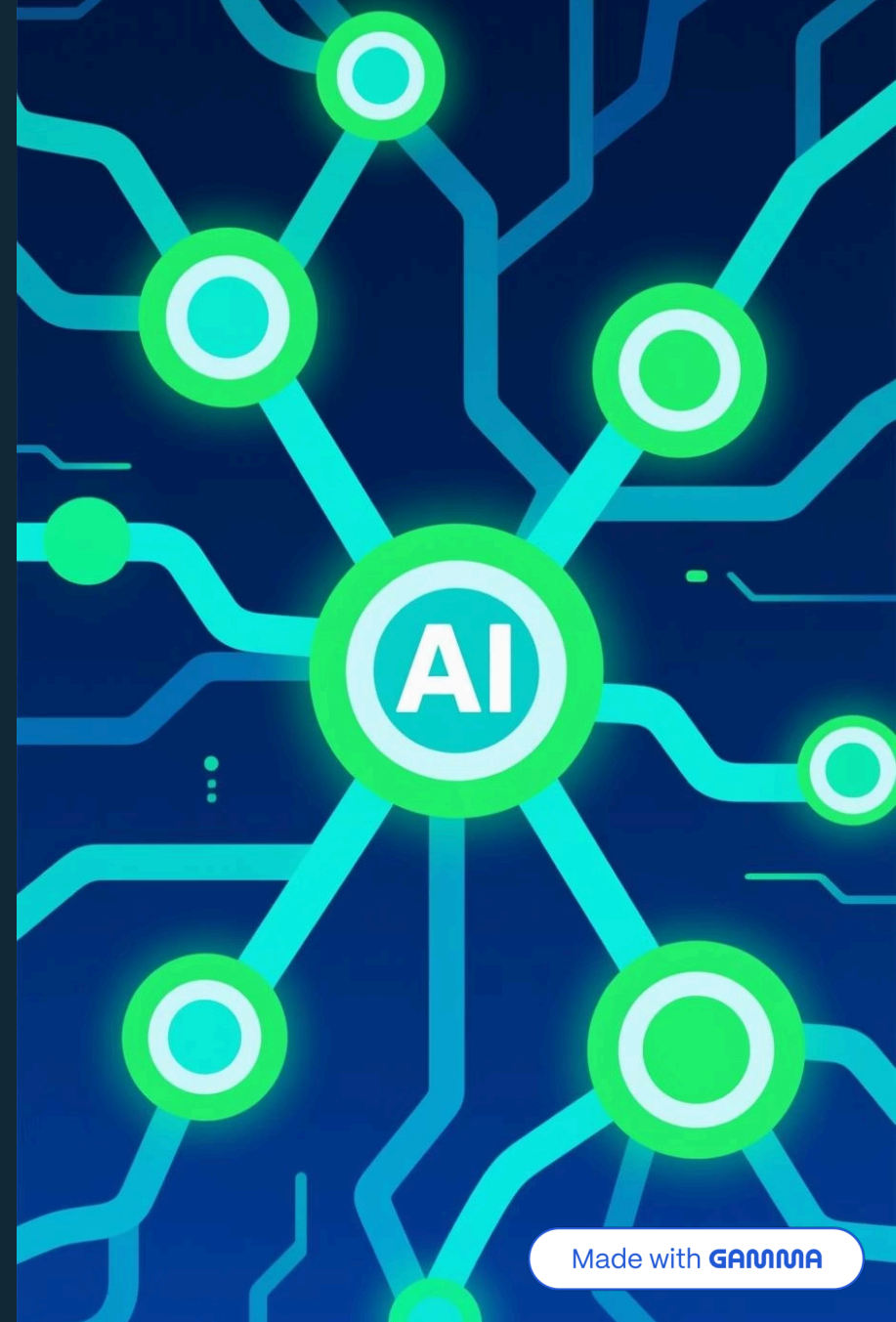


LLM Judge & Synthesiser Agent

Three minds, one superior answer — powered by intelligent evaluation and synthesis

Built with n8n + multi-LLM orchestration



Made with **GAMMA**



The Problem: Single LLMs Are Unreliable

Inconsistent Outputs

The same prompt to a single LLM can produce wildly different responses across attempts, making reliability unpredictable.

Model Bias & Hallucinations

Every LLM carries inherent biases from training data and tends to confidently present false information as fact.

Blind Selection

Current systems either pick responses randomly, average them poorly, or trust a single model without validation.

Why This Problem Has Real Consequences



High-Stakes Domains Pay the Price

In finance, a hallucinated number can trigger wrong investment decisions. In legal contexts, incorrect precedent citations damage cases. In education, students learn false information. In operations, bad recommendations waste resources.

Low-quality answers aren't just annoying—they carry real costs in credibility, money, and trust.



Our Solution: Judge *Before* You Generate

1

Multi-LLM Input

Query three different LLMs simultaneously to capture diverse reasoning approaches and knowledge bases.

2

Intelligent Judge

Dedicated agent evaluates all responses across accuracy, relevance, clarity, and usefulness—no blind trust.

3

Synthesised Output

Combine the best elements from all responses into a single, refined, higher-quality answer.

We prioritise quality over speed—evaluation happens before generation, not as an afterthought.

System Architecture Overview

O1

User Query Received

Single input triggers the entire workflow through n8n orchestration layer.

O2

Parallel LLM Calls

Three different LLMs (GPT-4, Claude, Gemini) process the query simultaneously without bottlenecks.

O3

Judge Agent Evaluation

Dedicated judge agent scores and ranks all responses using structured reasoning criteria.

O4

Synthesis Step

Best elements from top responses are merged into a coherent, refined final answer.

O5

Quality Output Delivered

User receives a single, superior response backed by multi-model validation.

How the Judge Agent Actually Works

Evaluation Dimensions

- **Accuracy:** Factual correctness and verifiability
- **Relevance:** Directly addresses the user's query
- **Clarity:** Easy to understand, well-structured
- **Usefulness:** Actionable and practical value



The judge assigns weighted scores across dimensions, ranks responses, and identifies specific strengths to preserve during synthesis. This isn't popularity voting—it's structured, bias-resistant decision-making that can detect when all responses are weak.

What Makes This Non-Trivial

Not Majority Voting

We don't count identical answers. The judge evaluates reasoning quality, not consensus.

Not Simple Summarisation

We don't mash responses together. We critically assess, then reconstruct from the strongest elements.

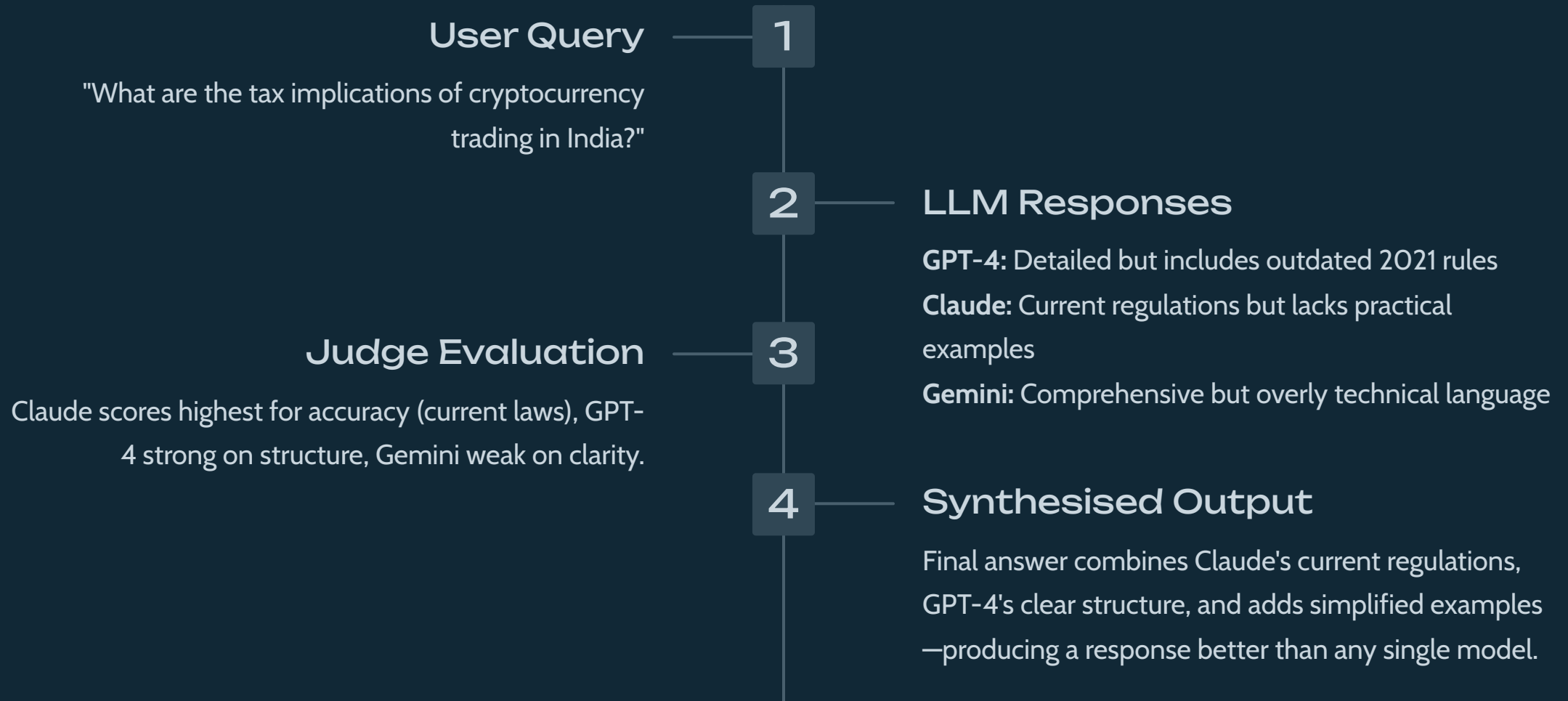
Structured Reasoning + Critique

The judge provides explicit reasoning for its decisions, making the evaluation process transparent and auditable.

Failure Handling Built-In

When all responses are weak, the system acknowledges limitations rather than forcing a poor synthesis.

Live Example: From Three to One





Why We Built This on n8n



Visual Orchestration

Complex logic flows become clear and maintainable through n8n's intuitive node-based interface.



Modular Agents

Each agent—LLM caller, judge, synthesiser—is an independent, reusable component that can be modified without breaking the system.



Easy Extensibility

Adding new LLMs, evaluation criteria, or synthesis strategies takes minutes, not days of refactoring code.



Production-Ready

Built-in error handling, monitoring, and scaling capabilities mean this isn't just a demo—it's deployment-ready.

Why This Beats Existing Approaches

vs. Single LLM

We eliminate single-point-of-failure risks. One model's hallucination or bias is caught by cross-validation from other models, dramatically improving reliability.

vs. RAG-Only Systems

RAG improves retrieval but doesn't validate generation quality. We add a critical evaluation layer that RAG lacks, ensuring retrieved information is used intelligently.

vs. Ensemble Averaging

Averaging dilutes good answers with bad ones. Our judge actively selects and synthesises the best elements, preserving quality instead of compromising it.